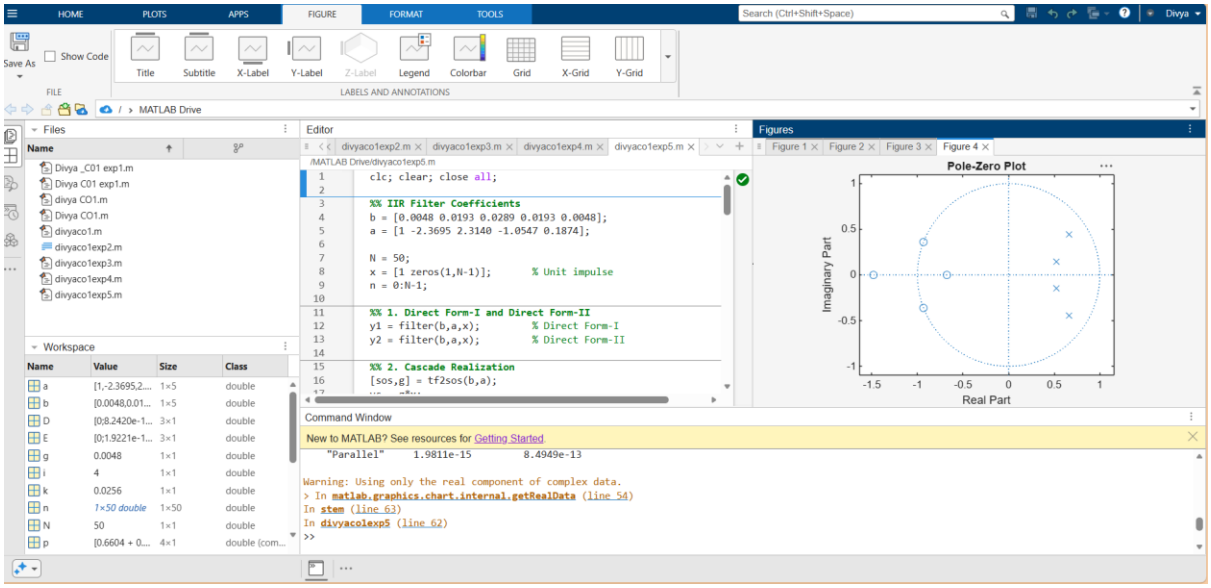
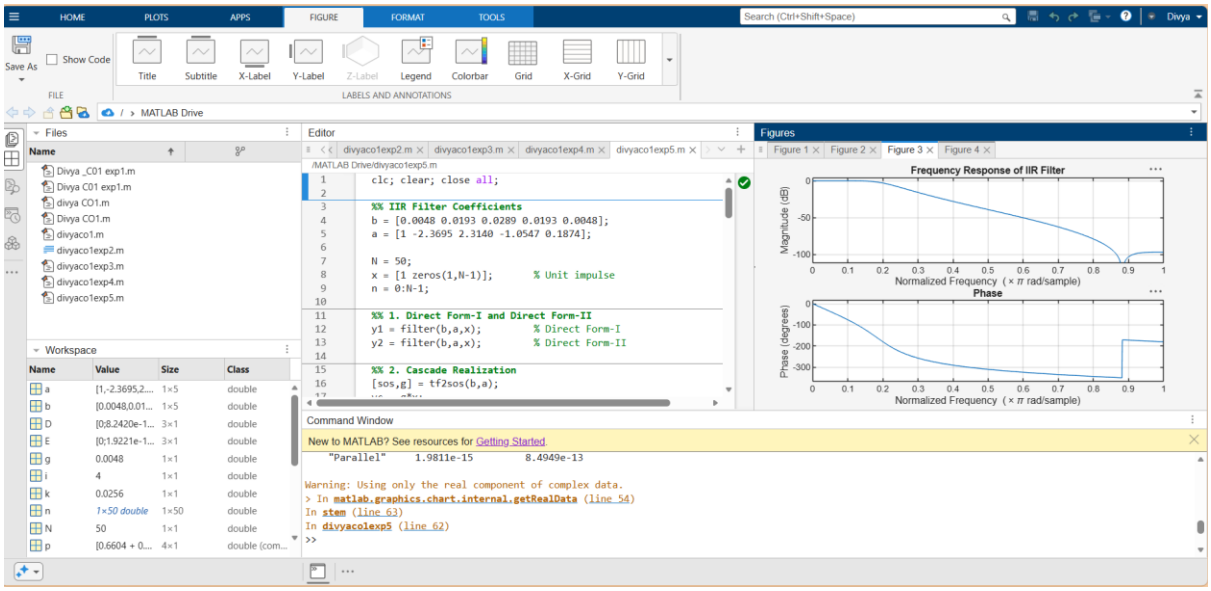
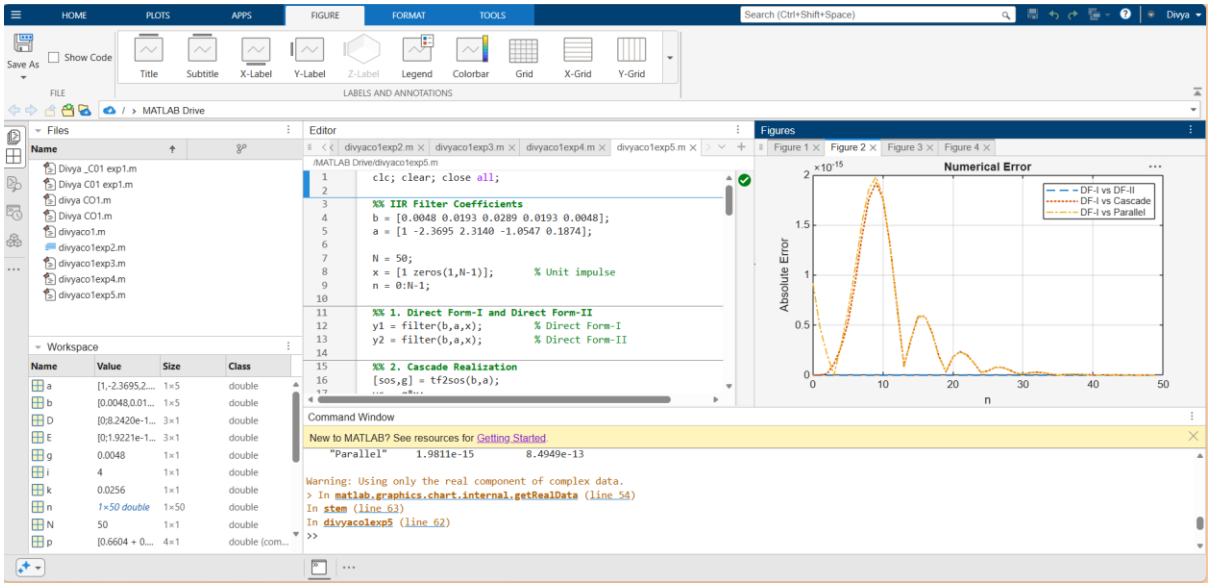
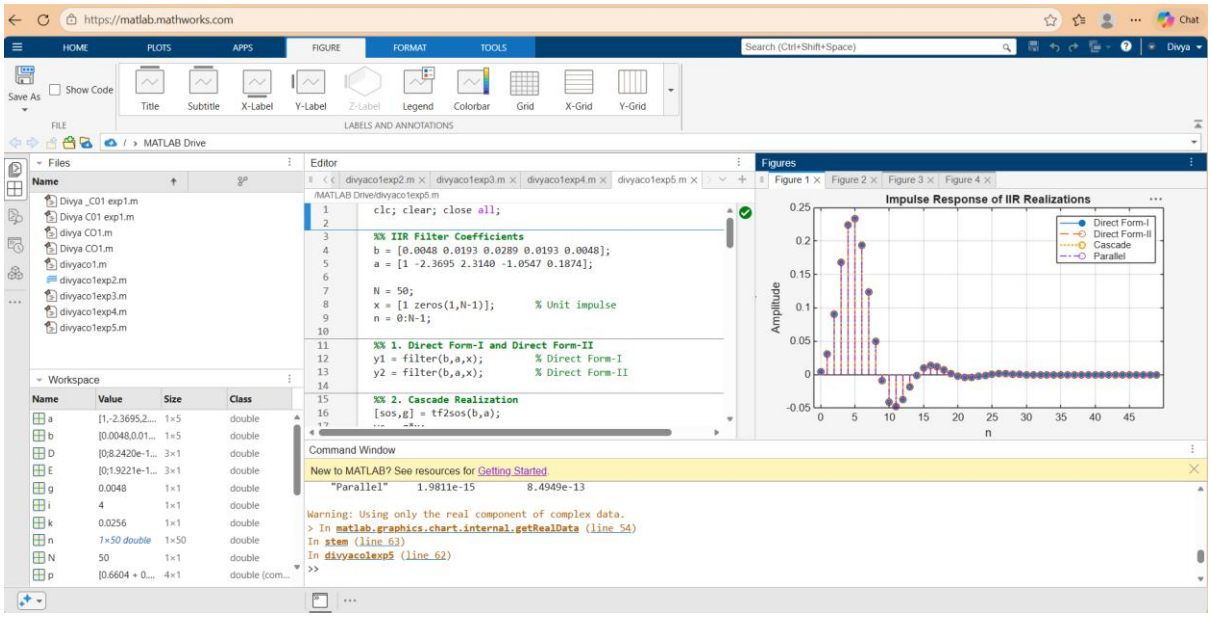

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divya CO1.m
Divya CO1.m
divyaco1.m
divyaco1exp2.m
divyaco1exp3.m
divyaco1exp4.m
divyaco1exp5.m
Workspace
Name
Value
Size
Class
Editor
divyaco1exp2.m
divyaco1exp3.m
divyaco1exp4.m
divyaco1exp5.m
/MATLAB Drive/divyaco1exp5.m
clc; clear; close all;
%% IIR Filter Coefficients
b = [0.0048 0.0193 0.0289 0.0193 0.0048];
a = [1 -2.3695 2.3140 -1.0547 0.1874];
N = 50;
x = [1 zeros(1,N-1)];
% Unit impulse
n = 0:N-1;
%% 1. Direct Form-I and Direct Form-II
y1 = filter(b,a,x);
% Direct Form-I
y2 = filter(b,a,x);
% Direct Form-II
%% 2. Cascade Realization
[sos,g] = tf2sos(b,a);
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Figure 4
Pole-Zero Plot
Imaginary Part
Real Part
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Warning: Using only the real component of complex data.
> In matlab.graphics.chart.internal.getRealData (line 54)
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Figures
Figure 1
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Figure 4
Frequency Response of IIR Filter
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)
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divyaco1exp4.m
divyaco1exp5.m
Workspace
Name
Value
Size
Class
a
[1,-2.3695,2....
1×5
double
b
[0.0048,0.01...
1×5
double
D
[0;8.2420e-1...
3×1
double
E
[0;1.9221e-1...
3×1
double
g
0.0048
1×1
double
i
4
1×1
double
k
0.0256
1×1
double
n
1×50 double
1×50
double
N
50
1×1
double
p
[0.6604 + 0....
4×1
double (com...
Editor
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% Direct Form-II
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Numerical Error
DF-I vs DF-II
DF-I vs Cascade
DF-I vs Parallel
Absolute Error
n
Command Window
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k
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n
1×50 double
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double
N
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Figure 2
Figure 3
Figure 4
Impulse Response of IIR Realizations
Direct Form-I
Direct Form-II
Cascade
Parallel
Amplitude
n
Command Window
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